CAMBRIDGE O LEVEL COMPUTER SCIENCE

SECONDARY 2 COMPUTER SCIENCE · CLASS TEST 1 MASTER REVISION GUIDE

Syllabus Scope: Sections 1.1 to 1.1.3 (Number Systems, Binary, Denary, Hexadecimal & Practical Uses)

1. Understanding the Binary System (Section 1.1.1)

Why do computers process data in Binary (0s and 1s)?

• Computers are built from microprocessors containing millions of tiny electronic switches called **transistors** and **logic gates**.

• Electrical circuits operate with only **two stable physical states**: **ON** (voltage present / represented by binary **1**) and **OFF** (no voltage / represented by binary **0**).

• Using binary prevents electrical noise, signal ambiguity, and provides reliable storage and data processing.

2. Binary ↔ Denary Conversions (Section 1.1.2)

Mastering Place Values: Every position in a binary number has a place weighting of 2^n.

Place Value (2^n)	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Denary Weight	**128**	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Binary Example (174)	**1**	**0**	**1**	**0**	**1**	**1**	**1**	**0**

Conversion Methods:

• **Binary → Denary:** Add up the denary weights for every column containing a '1'. (e.g. 01101011 = $64+32+8+2+1 = 107$).

• **Denary → Binary (Successive Division by 2):** Divide repeatedly by 2, write remainders (0 or 1), and read from **bottom to top**.

Example for 174: $174/2 = 87$ r 0; $87/2 = 43$ r 1; $43/2 = 21$ r 1; $21/2 = 10$ r 1; $10/2 = 5$ r 0; $5/2 = 2$ r 1; $2/2 = 1$ r 0; $1/2 = 0$ r 1 → **10101110**.

3. The Hexadecimal System (Base 16) (Section 1.1.2)

Hexadecimal uses 16 digits: **0 to 9** and letters **A (10), B (11), C (12), D (13), E (14), F (15)**.

Denary	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
4-Bit Binary	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111

Conversion Rules:

• **Binary → Hex:** Group the binary number into **4-bit nibbles from the right**. Convert each nibble to its hex digit.

Example: 0111 1101 0100 → 0111 (7) | 1101 (D) | 0100 (4) = **7D4**.

• **Hex → Binary:** Expand each hex digit into its exact **4-bit binary equivalent**.

Example: 5B8 → 5 = 0101, B = 1011, 8 = 1000 → **010110111000**.

• **Hex → Denary:** Multiply by powers of 16 ($16^2=256, 16^1=16, 16^0=1$).

Example for 2C5: $(2 \times 256) + (12 \times 16) + (5 \times 1) = 512 + 192 + 5 =$ **709**.

• **Denary → Hex:** Successive division by 16; write remainders as hex digits from bottom to top.

Example for 490: $490/16 = 30$ r 10 (A); $30/16 = 1$ r 14 (E); $1/16 = 0$ r 1 → **1EA**.

4. Practical Uses of Hexadecimal (Section 1.1.3)

Why use Hexadecimal instead of Binary?

1. **More Compact:** 1 Hex digit replaces 4 binary bits, saving screen display space.
2. **Human-Friendly:** Much easier for humans to read, write, and remember, drastically reducing transcription errors when debugging.

Application	How Hexadecimal is Used & Structure	Exam Example
MAC Address	Unique 48-bit (6-byte / 12 hex digit) hardware address hardcoded on a NIC. • First 6 hex digits: Manufacturer Code (OUI) • Last 6 hex digits: Device Serial Number • UAA (Permanent) vs LAA (Locally changeable by admin)	`4A:5B:6C:7D:8E:9F`
IPv6 Addresses	128-bit network address written as 8 groups of 4 hexadecimal digits separated by colons.	`2001:0db8:85a3::8a2e:0370:7334`
HTML Color Codes	6-digit hex code represents RGB color intensity (0–255 per channel). • #RR (Red), #GG (Green), #BB (Blue) • #000000 = Black, #FFFFFF = White, #FF0000 = Red	`#4A00E0` `(R=74, G=0, B=224)`
Memory Dumps	Outputs raw RAM contents during software crashes to help programmers inspect variable states and pinpoint faults.	`Address: 0x00401A` `Data: 4E 6F 74 20 46`
Error Codes & Assembly	Hex numbers identify specific hardware driver/memory errors and display assembly language opcodes and addresses.	`0x80070057`

5. Top Exam Tips & Pitfalls to Avoid

✔ Do's for Full Marks	✘ Common Traps to Avoid
• Always write full 8 bits (add leading zeros, e.g. 00101001). • Show all working steps (division remainders and multiplication powers) for method marks. • Memorize powers of 16: $16^0=1, 16^1=16, 16^2=256, 16^3=4096$.	• Do NOT confuse Hex letters: A=10, B=11, C=12, D=13, E=14, F=15. • Do NOT group binary from the left for Hex; always group in 4s from the **RIGHT**. • In MAC address, first 6 hex digits are Manufacturer (OUI), NOT device serial.